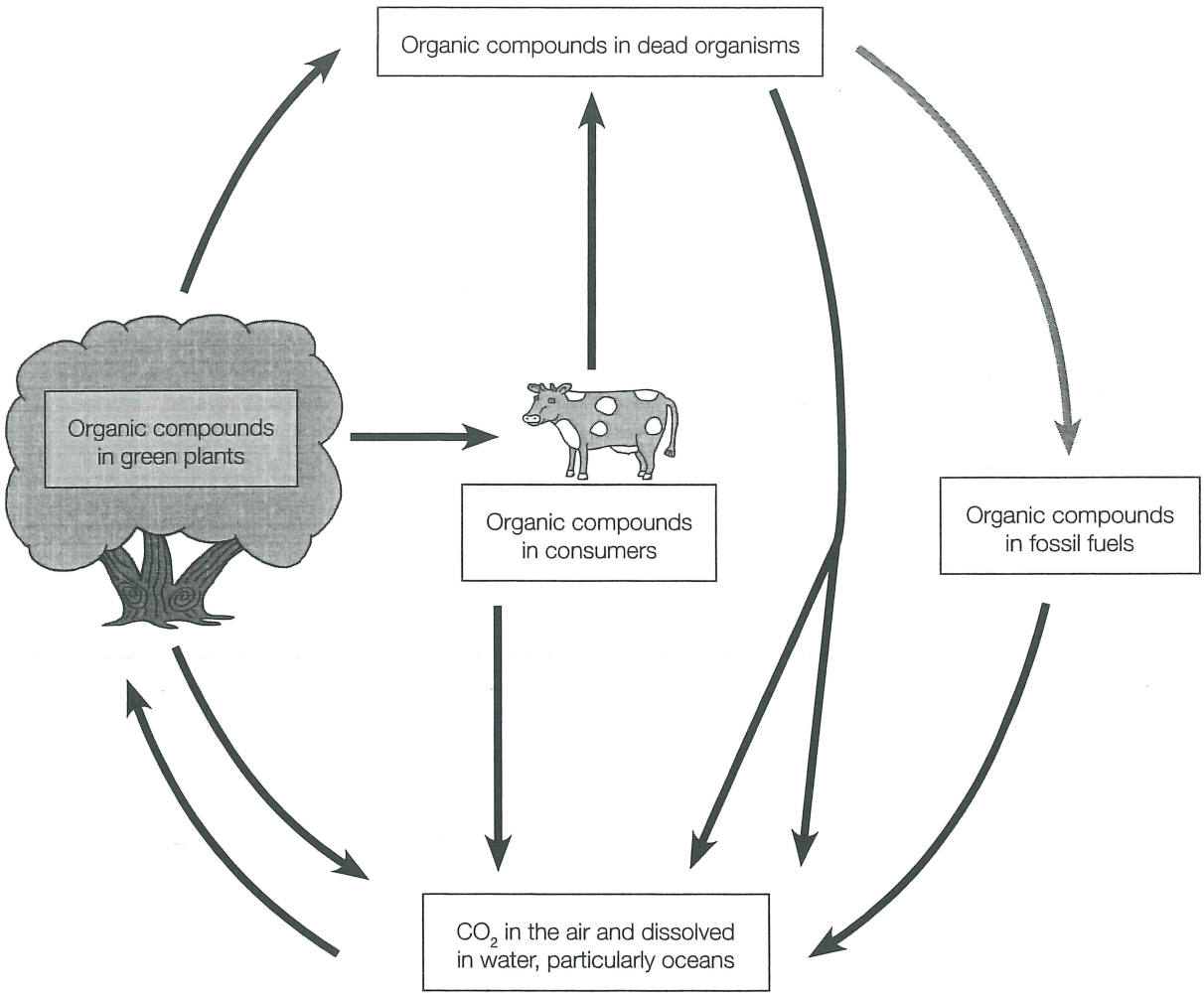
Carbon cycle

1. Use the words from the box to label the black arrows in the diagram below.

Photosynthesis removes CO_2 from ...			Respiration returns CO_2 to the ...	
Decay and decomposition – CO_2 released as microbes				
Death	Death	Feeding	Burning	Respiration



Answer the rest of the questions in your book.

2. Describe why carbon is so important to living things.

3. Describe the main sources of carbon on Earth.

4. Explain why decomposers are used in situations such as sewage treatment plants and in compost heaps.

5. Explain why plants are both consumers and producers of carbon. Refer to respiration and photosynthesis in your answer.

6. Explain how the environment is affected by humans continuing to burn fossil fuels.

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The greenhouse effect

1. Fill the gaps to complete this passage.

   The atmosphere is made up of three layers: the (a)_____, (b)_____ and (c)_____

   The heat from the Sun travels through space to earth by (d)_____ While in the atmosphere, infrared radiation can be absorbed by particles in the atmosphere such as (e)_____ and (f)_____ and it cannot be reflected back out into space. The result can be global (g)_____

2. Describe the greenhouse effect.

3. State three greenhouse gases.

4. Describe what greenhouse gases do to the heat energy from the Sun compared with non-greenhouse gas particles that are also in the atmosphere.

5. In your book, explain why scientists say that the Earth's atmosphere is like glass in a greenhouse.

6. In summer, weather forecasts often give information on the UV index. What are they referring to?

7. Explain why rising carbon levels in the atmosphere worry scientists in relation to greenhouse gases.

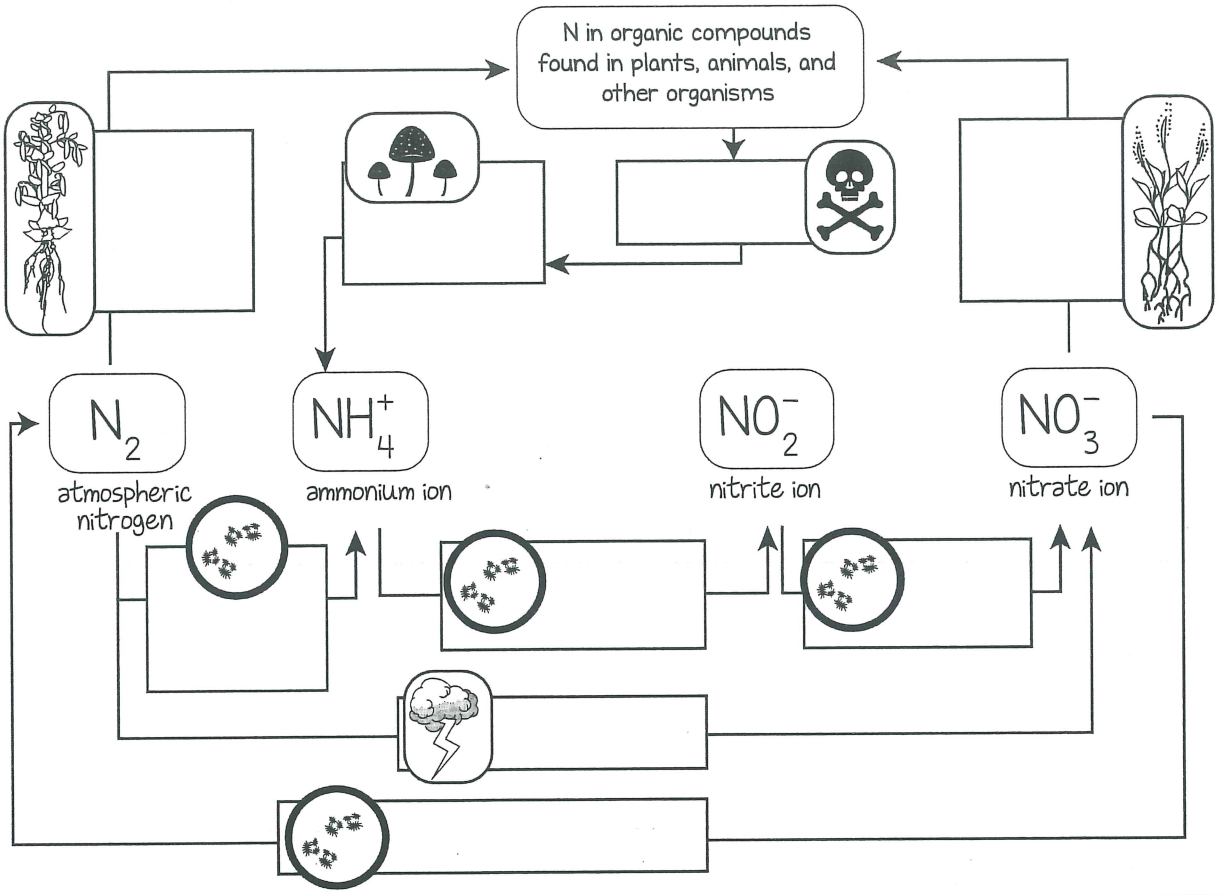
8. Describe the link between rising greenhouse gases in the atmosphere and rising sea levels.

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Name: Class: Date:

Nitrogen Cycle

Using the wordbank below, fill in the boxes to describe the players in the nitrogen cycle



1. decomposition by fungi and bacteria
2. lightning
3. most plant uptake
4. dead organisms and waste
5. nitrogen fixing bacteria in legume root nodules
6. nitrogen fixing bacteria in soil
7. denitrifying bacteria
8. ammonium oxidizing bacteria
9. nitrite oxidizing bacteria

Nitrogen is found in abundance in the atmosphere as N2, but this gas is not biologically available to organisms until it is "fixed" into ammonium ions. Nitrogen available to organisms to use is often the limiting ingredient, the limiting factor that determines plant productivity in an ecosystem.

10. When farmers add fertilizer to soil, what are they normally adding to the soil? What form(s) of nitrogen? (look it up if you don't know!)

_____

_____

Much of the nitrogen that humans add to the soil ends up in our streams, rivers, and the ocean. An increase of nitrogen in the waters can change the pH of the water, it can cause lower levels of oxygen in the water, and it can affect which organisms live there.

11. One consequence of increased aquatic nitrogen is toxic algal blooms. What is an algal bloom?

_____

_____

12. How do algal blooms affect other life in the surrounding area?

_____

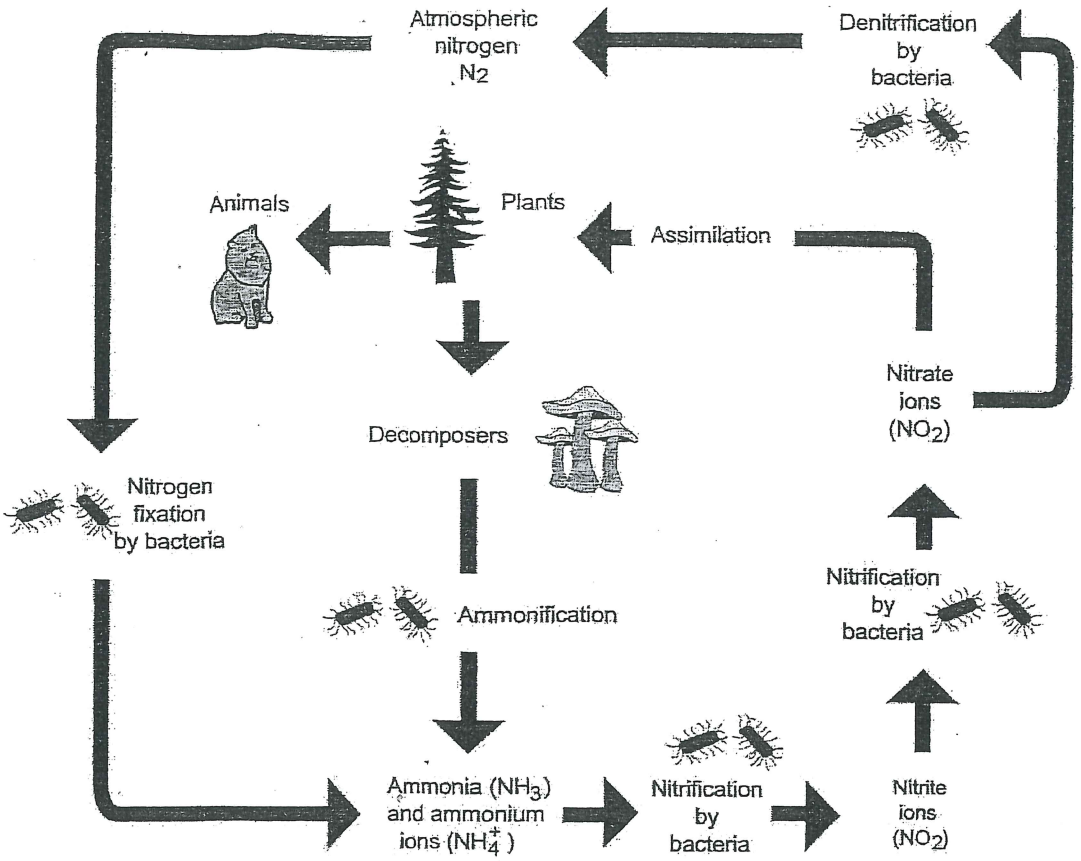
_____

_____

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Nitrogen Cycle -

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Nitrogen makes up _____% of the atmosphere. It is required for the production of _____ and _____.

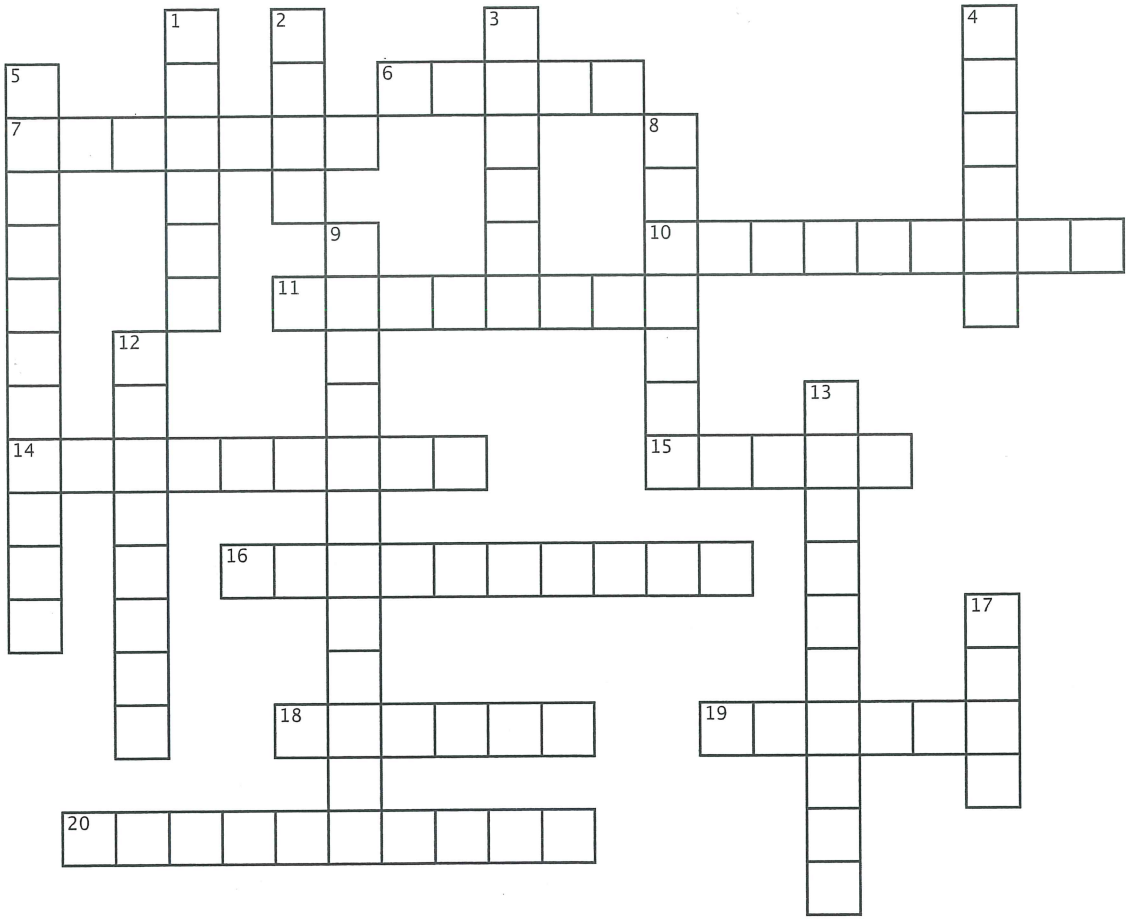
Steps of Nitrogen Cycle:

1. Nitrogen Fixation: conversion of nitrogen in the air to _____. Done in 2 ways: a) _____ b) _____.
2. Nitrates in soil taken up by _____
3. _____ consume these and use the amino acids to make _____.
4. Organisms produce waste and die. _____ break down nitrogen compounds into _____.
5. _____ is converted to nitrites, then to _____ by _____ in the soil.
6. Denitrification: Nitrates > nitrites > _____ released to air. Done by _____.

Human Impact: Agriculture (answer on back)

What is the impact of excess fertilizers on soils? What is the impact of run off of fertilizer into bodies of water?

HUMAN IMPACT



ACROSS

6. A source of electrical energy derived from the sun
7. Word for non-living
10. A strategy used by waste managers
11. Essential element in protein manufacture
14. The contamination of soil, water or the atmosphere by the discharge of harmful substances
15. Abiotic substances that make up the atmosphere
16. Atmospheric pollution enhances this effect
18. Name of gas extracted from waste
19. Plants are an essential part of this cycle
20. Arrows in a food chain, or food web, from one organism to another indicate ___

DOWN

1. Living
2. Type of rain that is produced by excessive amounts of nitrogen, carbon and sulfur oxides in the air
3. Name of a warming, circulation pattern that causes drought in Australia
4. Type of fuel emission from industry and motor vehicles
5. Abundant greenhouse gas
8. Method of hunting and agriculture as used by Aboriginals
9. Logging, housing and agriculture can reduce this, if not managed properly
12. Deep-rooting plants can alleviate this problem
13. Heat energy derived from the Earth
17. Name of energy harnessed from differences in atmospheric pressure

WORD LIST

- abiotic
- nitrogen
- gases
- pollution
- geothermal
- salinity
- fossil
- carbon
- solar
- biogas
- energy flow
- greenhouse
- biotic
- water vapour
- El Nino
- biodiversity
- acid
- wind
- burning
- recycling

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Worksheet 39: The ozone layer

Code cracker: Decode the symbols to find the missing words. Write the code you uncover in the table at the bottom of the page.

The ⊙♧⊙❀❖ layer is a layer of ozone ❁⊙◷❖☪†◷❖✝ that begins in the ✝✣卌⏃✣⊙✝✱✟❖卌❖ about ✣✟⑨卌✣❖❖❀ kilometres above Earth's surface. †◷✣卌⏃✵⑨⊙◷❖✣ light is much more energetic that the ◷⑨ℌ✟✣ and ✟❖⏃✣ energy from the Sun and it is much more damaging for us, for example by causing ✝†❀♧†卌❀ and skin ☪⏃❀☪❖卌.

An ozone molecule is made up of ✣✟卌❖❖ oxygen atoms covalently bonded together. The ozone layer's ✣✟⑨☪☢❀❖✝✝ varies depending on the season, ℌ❖⊙ℌ卌⏃✱✟⑨☪ position and which other molecules are in the atmosphere. Because the atmosphere ☪⑨卌☪†◷⏃✣❖✝ , the ozone layer is ✣✟⑨❀❀❖卌 at the ❖◷†⏃✣⊙卌 and poles. It is at its thinnest in ✝✱卌⑨❀ℌ over ⏃❀✣⏃卌☪✣⑨☪⏃ when a hole forms in the ozone layer and the ☪⊙❀☪❖❀✣卌⏃✣⑨⊙❀ of ozone molecules is so low that they cannot effectively ⏃♧✝⊙卌♧ the ultraviolet radiation from the Sun. The ozone hole is why there is such a high rate of skin cancers in ⏃†✝✣卌⏃◷⑨⏃ and New Zealand, and a lower rate in the United Kingdom and other countries that are further away from it. The ozone layer absorbs about 97 to 99 per cent of the Sun's radiation, mostly †◷✣卌⏃✵⑨⊙◷❖✣ light.

⊙♧⊙❀❖ is created when ultraviolet ◷⑨ℌ✟✣ hits an oxygen molecule (O_2) and ✝✱◷⑨✣✝ it into two oxygen ⏃✣⊙❁✝, which then combine with other ⊙Ψ♧ℌ❖❀ molecules in the atmosphere to create ozone molecules. These †❀✝✣⏃♧◷❖ molecules split into an oxygen ❁⊙◷❖☪†◷❖ and oxygen ⏃✣⊙❁ when more ultraviolet light hits them. The ultraviolet radiation is absorbed when it splits the molecules and is prevented from reaching Earth's ✝†卌☯⏃☪❖. As there is only a small amount of ozone in the ✝✣卌⏃✣⊙✝✱✟❖卌❖, the ozone layer exists in a delicate ♧⏃◷⏃❀☪❖ and we need to be careful that we do not do anything to endanger it.

Some molecules can ⚢❖✝✣卌⊙♧ ozone in the atmosphere. The ozone ⚢❖✱◷❖✣⑨❀ℌ molecules that are best known are the ☪✟◷⊙卌⊙☯◷†⊙卌⊙☪⏃卌♧⊙❀✝ (CFCs), but there are others including ❀⑨✣卌⊙ℌ❖❀ oxides, NO and NO_2. These molecules, which are generated by humans, ☯◷⊙⏃✣ up to the stratosphere where they 卌❖⏃☪✣ with the ozone molecules. CFCs are also called ☯卌❖⊙❀✝ and were used extensively as refrigerants, ✱卌⊙✱❖◷◷⏃❀✣✝ and ✝⊙◷✵❖❀✣✝ throughout the 20th century. Now, in an effort to protect the ozone layer, there are extensive 卌❖ℌ†◷⏃✣⑨⊙❀✝ controlling how many CFCs can be produced.

What letter does each symbol represent?

⊙	ℌ	✝	♧	⏃	♧	♧	卌	⚢	☢	✵	☯	✟	†	✱	⑨	❀	◷	❖	◷	Ψ	☪	❁	✣

Cracker clue

卌	❖	⏃	☪	✣
R	E	A	C	T

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